

IBM Telco Customer Churn Prediction

Capstone Project

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Introduction

Why Churn Prediction Matters

Customer churn — when a subscriber cancels their service — is one of the biggest cost drivers in the telecom industry.

Industry insight: Acquiring a new customer costs way more than retaining an existing one. Even a 1% reduction in churn can translate into millions in saved revenue.

Goal: Build a predictive model that identifies at-risk customers before they leave, enabling targeted retention campaigns.

Project Objectives

- Perform comprehensive EDA on the IBM Telco dataset
- Build & compare 3 ML models
- Identify the top drivers of churn via SHAP analysis
- Segment customers into risk tiers
- Visualize findings in a Tableau dashboard

Problem Description

7,043

Total Customers

IBM Telco dataset

26.5%

Churn Rate

Target variable

21

Features

Demographics + account + services

Dataset Feature Groups

Demographics

Gender, Senior Citizen, Partner, Dependents

Account Info

Tenure, Contract type, Payment method, Paperless billing

Services

Phone, Internet, Streaming, Security, Tech support

Charges

Monthly charges, Total charges

Methodology & Preprocessing

1

EDA

- Explored churn by contract, tenure & service type



2

Preprocessing

- Imputed blanks, OneHotEncoder + StandardScaler
- Addressed missing and blank values
- 80/20 stratified train/test split



3

Model Implementation & Comparison

- Logistic Regression, Random Forest, Gradient Boosting
- 10-fold CV across AUC, F1, Precision & Recall



4

Risk Factor Creation

- Predicted churn probabilities on test set
- Segmented into High ($\geq 60\%$), Medium (30–60%), Low ($< 30\%$)



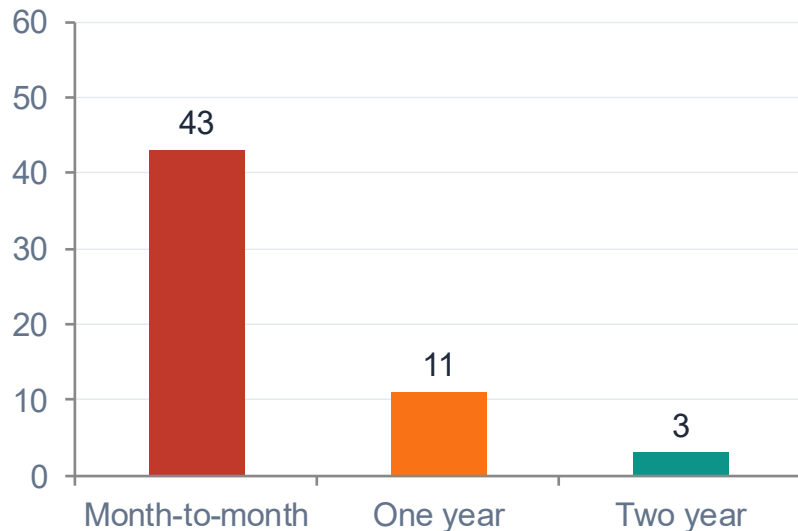
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Feature Importance / SHAP

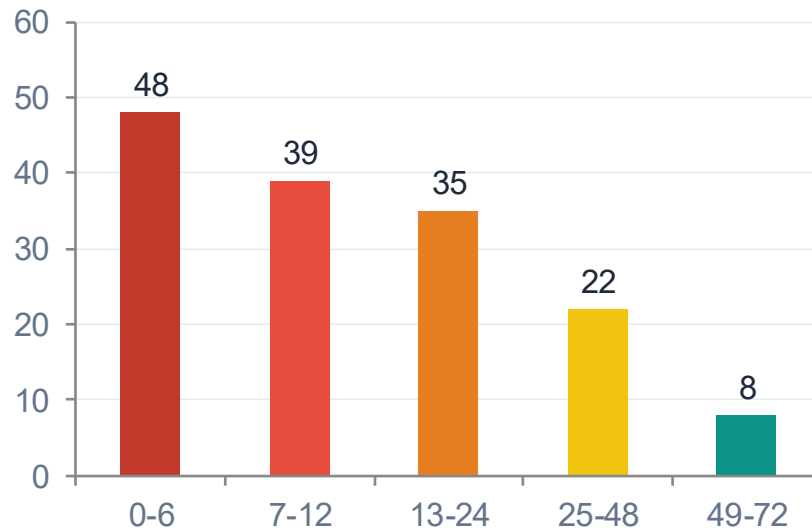
- Logistic Regression coefficients ranked
- SHAP LinearExplainer for directional insight
- Top drivers: Contract type, Tenure, InternetService

Exploratory Data Analysis — Key Insights

Churn Rate by Contract Type



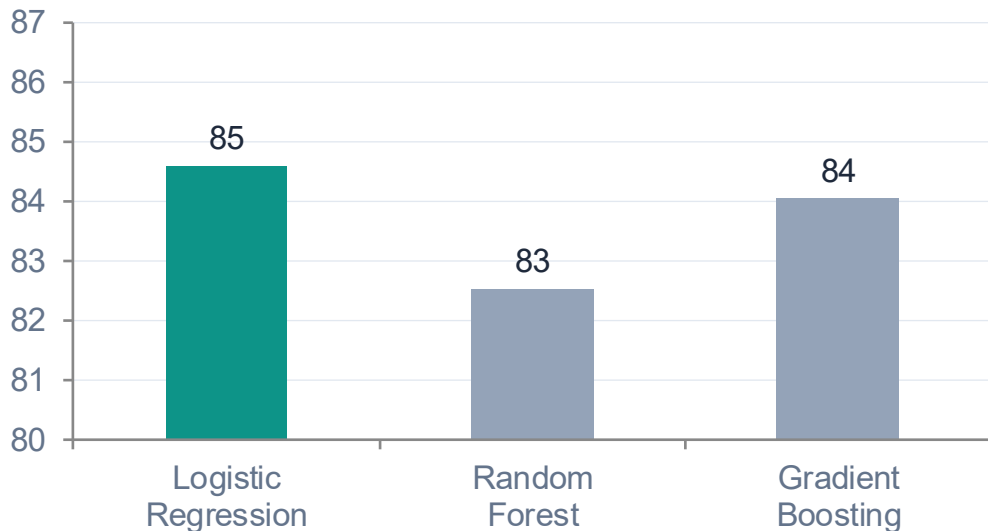
Churn Rate by Tenure Band (months)



- ▶ Month-to-month customers churn at 43% vs just 3% on two-year contracts
- ▶ New customers (0–6 months) have 48% churn; drops to 8% after year 4
- ▶ Fiber optic internet users churn at 43% vs 19% for DSL
- ▶ Higher monthly charges correlate with elevated churn probability

Model Development & Results

10-Fold Cross-Validated AUC (×100)



Performance Summary

Model	AUC	Accuracy
Logistic Reg ★	0.8459	80.3%
Random Forest	0.8253	78.9%
Gradient Boost	0.8403	79.8%

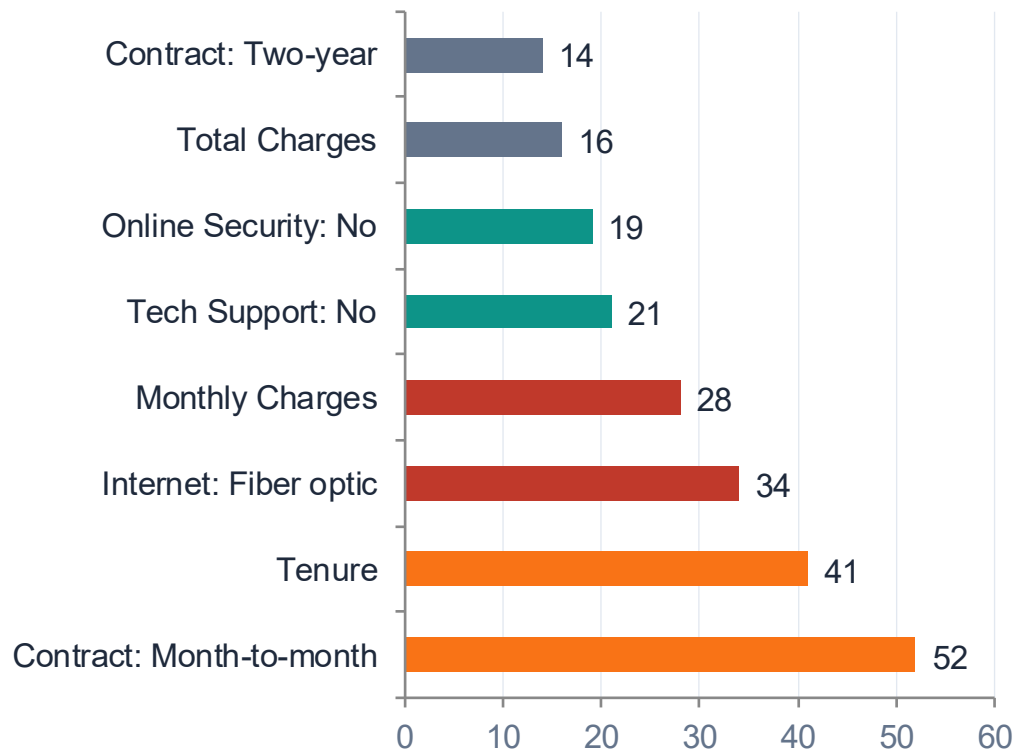
★ Winner: Logistic Regression

Highest AUC (0.846) with best accuracy. Differences between models are small (0.01–0.03), confirming logistic regression is a strong baseline for structured tabular churn data.

Cross-Validation (10-fold stratified): All models were evaluated using 5 metrics — ROC-AUC, Accuracy, F1, Precision & Recall — ensuring robust, unbiased performance estimates on unseen data.

Key Churn Drivers — SHAP & Feature Importance

Top Churn Drivers (Mean |SHAP| ×100)



Contract Type

Month-to-month = highest churn. Promoting annual contracts is the single biggest lever.

Tenure

Customers leave early. First 12 months are critical — focus onboarding & satisfaction here.

Internet Service

Fiber optic users churn more — possibly due to higher costs or unmet performance expectations.

Monthly Charges

Higher bills without perceived value drive churn. Bundled savings can help retention.

Tech Support

Customers without tech support churn more. Offering it as a free trial may improve retention.

Risk Segmentation & Business Application

HIGH RISK

≥ 60% churn prob.

15%

of customers (n=209)

Recommended Action

Immediate personal outreach, discount offers, contract upgrade incentives

MEDIUM RISK

30–60% churn prob.

24%

of customers (n=333)

Recommended Action

Targeted email campaigns, service upgrades, loyalty points, add-on trials

LOW RISK

< 30% churn prob.

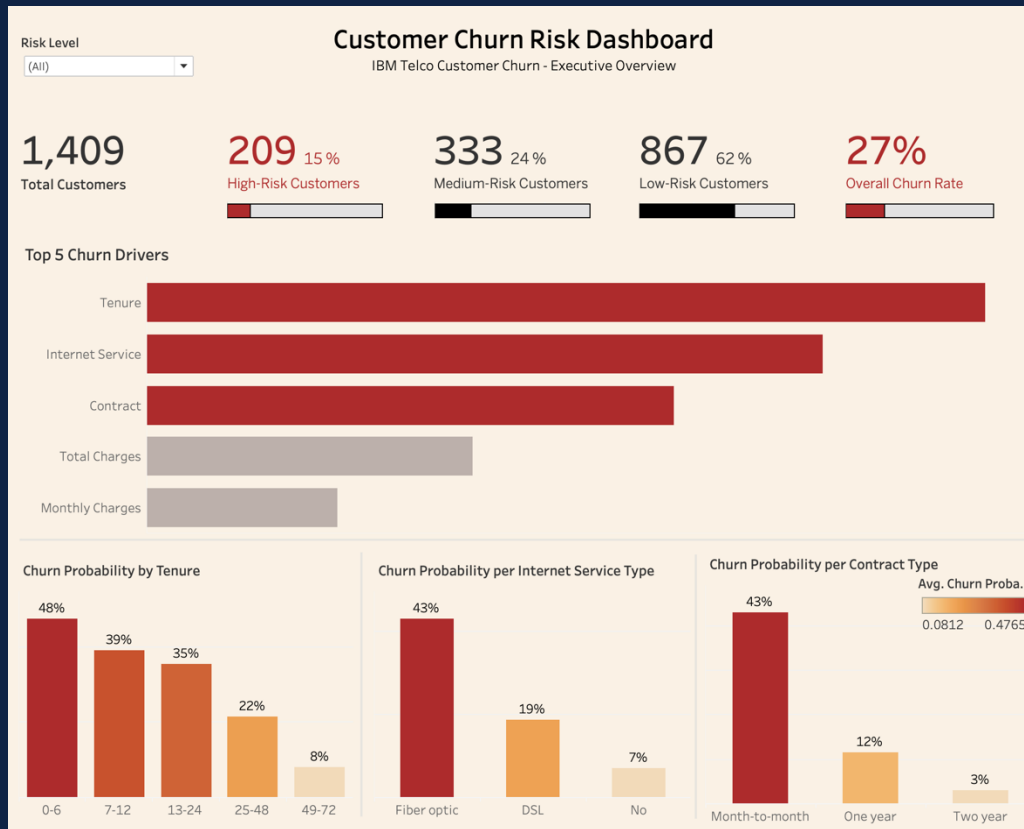
62%

of customers (n=867)

Recommended Action

Standard retention, upsell opportunities, satisfaction surveys

Live Dashboard Walkthrough



1,409 test-set customers scored • 209 High-Risk (15%) flagged for outreach • Top 5 churn drivers validated visually • Churn probability by tenure, internet type & contract

Conclusions

01

Logistic Regression achieved the best CV AUC (0.846), outperforming tree-based models on this tabular dataset.

04

SHAP analysis confirmed that model predictions are explainable and aligned with domain knowledge.

02

Contract type and customer tenure are the strongest predictors of churn — structural, actionable factors.

05

Risk tiering (High/Medium/Low) translates raw probabilities into clear, actionable business intelligence.

03

Fiber optic customers represent a high-churn, high-value segment requiring targeted quality & pricing strategies.

06

An asymmetric cost function ensures the model prioritizes recall over precision — appropriate for churn use cases.

References

Dataset

- IBM Telco Customer Churn — Kaggle ([blastchar/telco-customer-churn](#))
- Retrieved via kagglehub Python package

Thank You!

References

Dataset

- IBM Telco Customer Churn — Kaggle (blastchar/telco-customer-churn)
- Retrieved via kagglehub Python package

ML Libraries

- scikit-learn — Pedregosa et al. (2011). Scikit-learn: Machine Learning in Python. JMLR 12, 2825–2830.
- SHAP — Lundberg & Lee (2017). A Unified Approach to Interpreting Model Predictions. NeurIPS.

Visualization

- Tableau Desktop — Tableau Software (2024). tableau.com
- Matplotlib & Seaborn — Hunter (2007) / Waskom (2021)

Tools & Environment

- Python 3.10 | Jupyter Notebook | pandas | NumPy
- IBM Cloud / Watson Studio (project context)